

4. Answer both parts (a) and (b).

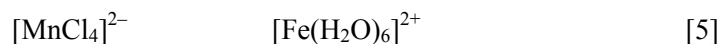
(a) Predict the structures of all of the following species using the valence shell electron pair repulsion (VSEPR) theory:



(b) For *THREE* of the species you have drawn above identify on your sketch (i) the highest order proper rotation axis, and (ii) one additional symmetry element. [5]

5. Answer both parts (a) and (b).

(a) Draw crystal field splitting diagrams for the metal 3d orbitals in the following complexes. Label the 3d orbital sub-sets and identify the number of unpaired electrons in each case:



(b) Account for the variation in the hydration energies (kJ mol^{-1}) of the following transition metal ions:

Mn^{2+}	Fe^{2+}	Co^{2+}	Ni^{2+}	Cu^{2+}	Zn^{2+}	
1814	1889	2023	2074	2069	2013	[5]

Turn over